Artur Mrozowski

Senior Data Architect / Analytics Engineering Lead

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Summary

Senior data architect and analytics engineering lead with 15+ years of experience designing source-of-truth data platforms, semantic data models, and operational analytics solutions across insurance, finance, and pharma.

Strong background in analytics strategy, data product platforms, metadata-driven automation, event-driven analytics, and cross-functional collaboration between business and engineering teams.

I specialize in turning complex business processes into governed analytical platforms that improve metric consistency, operational visibility, and decision support.

Experience

Senior Analytics Engineer, Tryg — Oct 2023 – present

- **Revenue Intelligence and Operational Analytics:** Developed a revenue intelligence platform enabling standardized business metrics including NDR, GRR, Logo Retention, and funnel analytics across domains.
- Built an event-driven analytical model enabling drill-down from executive KPIs to individual business events and operational decision points.
- Established reusable semantic definitions and time-series business events to improve metric consistency, explainability, and cross-functional trust in analytics.
- **Data Product Platform and Strategy:** Helped shape a data product strategy focused on reducing fragmentation, improving interoperability, and enabling scalable analytics delivery.
- Built a shared data product platform improving consistency, reuse, and delivery speed across analytical domains.
- Replaced dbt with SQLMesh to improve scalability, maintainability, and developer workflow for analytical data products.
- **Metadata-Driven Automation:** Designed a metadata-driven model generation approach turning declarative YAML definitions into governed SQLMesh models.
- Automated generation of analytical building blocks including hook, bridge, and event models from reusable metadata, reducing manual SQL development and improving structural consistency across domains.
- Modeled temporal validity, dependency-aware bridge logic, and reusable business keys as metadata instead of hand-coded model patterns.
- Used dependency graphs and topological ordering to generate analytical structures in the correct sequence, improving repeatability and reducing delivery risk.

Senior Data Architect, Lundbeck — Sept 2021 – Sept 2023

- Defined a common data layer strategy reducing fragmentation and improving interoperability across analytics and operational domains.
- Replaced a monolithic data lake approach with a flexible platform inspired by Data Mesh principles.
- Enabled scalable self-service analytics while maintaining governance and semantic consistency.
- Collaborated across technical and business domains to align analytical capabilities with organizational needs.
- **Technologies:** Snowflake, dbt, AWS.

Data Engineer / Data Architect, Tryg — Aug 2016 – Aug 2021

- Contributed to an underwriting data platform project, replacing Excel-based workflows with centralized governed data flows and shared business logic.
- Introduced Event Modeling to improve collaboration between business and engineering stakeholders.
- Designed enterprise semantic and canonical models based on ACORD to standardize concepts and improve interoperability across domains.
- Built event-driven write-back capabilities using GraphQL and Kafka, enabling applications to publish operational outcomes back into the analytical ecosystem.
- Automated logical and physical model generation using metadata-driven techniques, improving consistency and reducing manual effort.
- **Enterprise Canonical Data Platform:** Developed a semantic data platform enabling standardized interchange of business information across departments.
- Implemented graph-based semantic models in Neo4j aligned with the ACORD insurance ontology.
- Exposed semantic models through GraphQL APIs and integrated event streams using Kafka.
- **Metadata and Data Catalog:** Implemented metadata-driven lineage and business glossary capabilities using semantic technologies and Neo4j.
- Exposed metadata and lineage through Amundsen data catalog.

Data Architect, KMD — Nov 2015 – Aug 2016

- Designed target architecture and business proposal for a large-scale MS SQL Server data warehouse initiative.
- Participated in Data Governance Board activities defining enterprise data standards, policies, and governance principles.

Senior DW/BI Developer, Nordea — Aug 2014 – Nov 2015

- Led evaluation and proof of concept for data warehouse automation tooling using WhereScape RED.
- Improved reliability and reduced time-to-market for BI deliveries through automation and streamlined delivery workflows.

Earlier Career

- Swedish Agency for Marine and Water Management - BI Consultant, 2013-2014
- DR - Business Intelligence Consultant, 2012-2013
- Danske Bank - External Consultant, 2010-2012
- SimCorp - Software Developer, 2008-2010
- Copenhagen Energy - DBA, 2004-2008
- Quibus International AB - Software Developer, 2002-2004

Skills

Strategy: Analytics strategy, operational intelligence, data product platforms, source-of-truth data

Modeling: Semantic modeling, canonical modeling, governed metrics, event-driven analytics, operational modeling

Methods: Metadata-driven automation, domain-driven design, Event Modeling, Data Vault, HOOK methodology

Governance: Data governance, lineage, metadata management, business glossary, reusable semantic definitions

Technologies: Snowflake, SQLMesh, dbt, SQLGlot, Kafka, GraphQL, Neo4j, AWS, Azure

Certifications

Certified Scrum Product Owner: 2022

Data Mesh - Domain Oriented Data: 2022

Lead Enterprise Architecture Program on Azure - Microsoft: 2017

Data Modelling Master Class - Steve Hoberman: 2016

Data Vault Certified Data Modeler, DVCDM: 2014

CDMP Preparation: 2024

Education

Lund University - Bachelor of Science, System Analysis

- Faculty of Informatics.
- Key areas: database systems, data modeling, software architecture, UML, object-oriented design, Java programming, algorithms, and data structures.

Professional Development

- Knowledge Graph Academy - ongoing
- Claude Academy - Claude Certified Architect track, ongoing
- Apache Spark Workshop for Developers
- Distributed Computing with Spark
- Data Science Specialization - Johns Hopkins University
- Statistical Analysis with R
- Oracle BI EE
- C++ Programming
- SQL Server Administration
- Oracle Administration

Speaking and Open Source

- *Democratizing Data* - Tryg and Prisma, 2021
- *Building Semantic Data Hub with LPG, GraphQL and Kafka Streams*, 2019
- GraphQL and graph data modeling in Neo4j
- Azure Functions and machine learning
- XML parser for Danish motor register data